

Solution Requirements

Date	15 July 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Disaster response personnel, meteorological observers, and administrative staff must log in securely using OAuth 2.0 with forced Multi-Factor Authentication (MFA).	High
2	Authorization levels	The system must enforce Role-Based Access Control (RBAC): "Disaster Officers" can authorize/trigger warning dispatches, "Meteorological Observers" can ingest and flag sensor telemetry, and "Public" users are limited to read-only map feeds.	High
3	External interfaces	Must connect continuously to the National Meteorological Agency's REST APIs for radar data, ingest live sensor streams via MQTT, and integrate with cellular gateways (Twilio API) for emergency SMS broadcasts.	High
4	Transactions processing	The backend must reliably process high-volume sensor streams, execute spatial prediction pipelines, and queue public alerting tasks concurrently without dropping database packets.	Medium
5	Reporting	The system must auto-generate dynamic visual inundation heatmaps, predictive level charts, and downloadable historical disaster logs in PDF/CSV format for post-event auditing.	High
6	Business rules, etc.	Emergency public alerts must only auto-dispatch if the ML classification model establishes a flood probability confidence threshold greater than 85%; otherwise, manual authorization from a regional coordinator is required.	High
7	Compliance to laws or regulations	Must strictly comply with national disaster mitigation data standards and ensure geographic metadata handling complies with general regional data protection acts.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
8	Other	Automated sensor diagnostic check-ups must run hourly to immediately flag offline units and switch ingestion feeds to secondary localized stations.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system must compute and render predictive flooding models instantly once live sensor telemetry is processed.	Model inference & map generation completes in < 3 seconds.
2	Scalability	The system must seamlessly handle large user traffic spikes during active severe weather events without degradation.	Maintains service stability up to 100,000 concurrent active dashboard hits.
3	Security & Data Privacy	All telemetry data in-transit must be encrypted, and storage for sensitive admin parameters must have high standard encryptions.	End-to-end TLS 1.3 encryption for ingestion and AES-256 for database.
4	Reliability & Availability	The alerting framework must maintain constant functionality even if secondary database centers go offline.	Maintains 99.99% monthly system uptime with cross-region replication.
5	Usability & Accessibility	The interface must be usable by all emergency coordinators, featuring intuitive map navigation.	100% compliance with Web Content Accessibility Guidelines (WCAG) 2.1 AA.
6	Other (Maintainability)	The entire microservices framework must be containerized to allow fast localized setup and cloud independent porting.	Deployable on standard Kubernetes/Docker hosts in under 15 minutes.